

Supplementary Material S1

Per-run provenance log: schema and references

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1 Purpose

This document specifies the structure of the per-run provenance log `summary.json` that every training and evaluation run in this paper writes alongside its model checkpoint. The body of the paper refers to this document via the label *Supplementary Material S1* whenever a numerical claim in the main text or a table needs to be traced back to a specific training run.

The intent is twofold: (i) make every headline number in the paper auditable to a single canonical artefact under the companion repository [kanto-calibrated-spt-prior](#), and (ii) give downstream readers a stable schema against which they can run their own reproduction or extension experiments.

2 File layout

For every run reported in the paper, the companion repository contains a directory of the form

`data/provenance/<run_name>/`

with the following files:

- `summary.json` — the canonical provenance log (this document’s subject).
- `foundation_model.pt` — the trained PyTorch checkpoint (when the run is a model training run).
- `foundation_model.meta.json` — minimal identification metadata required to load the checkpoint.
- `calibration_chosen.json` — chosen conformal radii by nominal α (calibration runs only).
- `checkpoints/` — intermediate checkpoints (training runs only).
- `diagnostics.png` — auto-generated residual / calibration diagnostic figure.

3 Schema

The `summary.json` top-level keys are listed in Table 1. Types follow Python’s JSON convention: *string*, *integer*, *float*, *bool*, *list[T]*, *dict*.

Table 1: Top-level schema of `summary.json`.

Key	Type	Description
<code>run_name</code>	string	Canonical identifier, e.g. <code>kanto_full_6k_50ep_linear_rbf</code> . Mirrors the directory name.
<code>n_smoke</code>	integer	Total number of SPT measurement rows used in this run. For the operational best run this equals 495,725.
<code>n_features</code>	integer	Number of input features after enrichment (co-variates + one-hot regime, etc.).
<code>n_epochs</code>	integer	Number of training epochs.
<code>batch_size</code>	integer	Mini-batch size used by the SVGP ELBO.
<code>n_inducing</code>	integer	Number of inducing points used by the variational approximation.
<code>final_loss</code>	float	Final negative ELBO at the end of training (Gaussian likelihood) or the equivalent Student- <i>t</i> objective (heavy-tail variants).
<code>training_time_seconds</code>	float	Wall-clock training time on the recorded device.
<code>target_mean</code>	float	Mean of the raw N- <i>value</i> target used for standardisation; identical across all runs (<i>cf.</i> § 2 of the paper).
<code>target_std</code>	float	Standard deviation of the raw N- <i>value</i> target used for standardisation.
<code>target_transform</code>	string	One of <code>none</code> , <code>log1p</code> , <code>boxcox</code> ; controls the inverse transform used for diagnostic plotting.
<code>regime_one_hot</code>	bool	Whether the eight-way lithology regime was one-hot encoded.
<code>heteroscedastic_noise</code>	bool	Whether a learnable per-input noise head was attached (null result; see Appendix D of the paper).
<code>kernel_type</code>	string	One of <code>rbf</code> , <code>matern12</code> , <code>matern32</code> , <code>matern52</code> .
<code>mean_type</code>	string	One of <code>constant</code> , <code>linear</code> .
<code>inducing_init</code>	string	One of <code>random</code> , <code>kmeans</code> , <code>kmeans_stratified</code> .
<code>likelihood_type</code>	string	One of <code>gaussian</code> , <code>studentt</code> .
<code>studentt_df_init</code>	float <i>or</i> null	Initial degrees-of-freedom for the Student- <i>t</i> likelihood; null for Gaussian runs.
<code>spatial_kfold</code>	list[dict]	Per-fold evaluation metrics (Section 4 below).
<code>regime_distribution</code>	dict	Map of lithology regime name $\rightarrow$ row count in the training set. Should sum to <code>n_smoke</code> .
<code>device</code>	string	<code>cpu</code> , <code>mps</code> , or <code>cuda:<id></code> .

4 The `spatial_kfold` block

For every run that participates in spatial cross-validation, `summary.json["spatial_kfold"]` is a list of one dict per fold, each with the following keys:

- `fold`: integer fold index, 0 to $K - 1$.
- `n_train`: number of rows in the training portion.
- `n_test`: number of rows in the held-out test portion.

- **rmse**: per-fold RMSE in raw N -value units.
- **mae**: per-fold MAE in raw N -value units.
- **std_mean**: mean predicted standard deviation across test rows (used by the calibration diagnostic).

For the operational best run (`kanto_full_6k_50ep_linear_rbf`), the three fold entries sum to $n_{\text{test}} = 495,725$; this is the in-distribution training-fit diagnostic (*cf.* Section 5.3 of the paper). The proper- K -fold counterpart (where each fold is trained from scratch on the $K-1$ training side) is reported separately in `nested_kfold_summary.json`, with the headline values 11.510 random / 13.697 contig.

5 Cross-reference to the paper

Table 2 maps each headline numerical claim in the paper back to the canonical `summary.json` (or `nested_kfold_summary.json`) file in the companion repository.

Table 2: Headline numerical claims and their canonical provenance files.

Section / Table	Claim	Canonical file
Abstract, §5.11	RMSE 5.875 (training-fit)	<code>kanto_full_6k_50ep_linear_rbf/summary.json</code>
Table 3	RMSE 11.510 (proper random K -fold)	<code>kanto_full_6k_50ep_linear_rbf/nested_kfold_summary.json</code>
Table 3	CatBoost RMSE 9.660 random	<code>catboost_tuned/nested_kfold_summary.json</code>
Table 3	LightGBM RMSE 12.188 contig	<code>lightgbm_tuned/nested_kfold_summary.json</code>
Table 4	Buffered random RMSE 15.004 ($\sim 93\%$ train shrinkage)	<code>buffered_random_k3/summary.json</code>
Table 4	Buffered contig RMSE 13.575 ($\sim 16\%$ train shrinkage)	<code>buffered_contig_k3/summary.json</code>
§5.15	Hybrid-xfit contig RMSE 12.999	<code>hybrid_xfit_k5/summary.json</code>
§5.16	Hurdle contig RMSE 11.523	<code>hurdle_k3/summary.json</code>
§5.7	Cox PH C-index 0.544 contig	<code>cox_ph/summary.json</code>

6 Notes

- Every `summary.json` is written by the `train_kanto_smoke.py` entry point of the companion repository’s `code/` directory; manual edits are forbidden.
- `nested_kfold_summary.json` aggregates per-fold retrained models; it is a strict superset of `summary.json` (which records the train-on-all diagnostic only).
- A run that aborts mid-training writes `summary.json` only if at least one epoch completed; partial runs are excluded from the paper’s analysis.